Opera Numerorum
Machine Certification for GRH(X_0(143)) and BSD(J_0(143))

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CERTIFIED: GRH for X_0(143) | BSD rank(J_0(143)) = 1 | alpha_0 = 299+pi/10

Master Manifest SHA-256 (M1-M6 concatenated stdout)
`5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9`

SHA-256 Certified Chain

Mod	Claim	Stdout SHA-256	Status
M1	alpha_0 = 299+pi/10 (5000 dps)	63ef870a78766619327e99b68683bceff8c8ef9a525298756c77c8378fd2c291	**CERTIFIED**
M2	kappa bound (80-bit long double)	3716c7dbb32524074b8fffb65eea45069c8b568a31dc73706405116b84029a83	**CERTIFIED**
M3	CF pi/10: Q_5=226, bound=82829	e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044	**CERTIFIED**
M4	S_14: 14 primes, p_5 > 82829	b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a096c18ce2e9e2a19ed	**CERTIFIED**
M5	C(S_4) = 11.4221 > 2*sqrt(13)	9df98a3970acbb6942770a6cdd42fb21b009f9a5f45a222dd963e98ba4cb7a13	**CERTIFIED**
M6	genus(X_0(143))=13, Bost bound	ec9fa8c3aad478312c7e0d7373904dc3407eb5e9f4c19a011e3ca2ccb84da9fb	**CERTIFIED**
M7	Master manifest over M1-M6	5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9	**LOCKED**
M8	rank(H_13(L_w, J_0(143))) = g = 13	e2d70821cd66588cd715dfe37a44122130f88d15584738f5f64a02ff7f7b0002	**CERTIFIED**
M8C	Z=15, M*=4/55, 200 Hodge classes	02fe604876c3253ec61ce0a8b382c7b01a089d1d217ab200fc9975464a645323	**ARCH.CERT**
M8J	delta=1.89m; tidal<0.1g; OQs closed	298d440aae8ecc3808b413c7ce1b1cf19c92d359beb7664d837062e04b01b505	**ARCH.CERT**
M8K	FTL Morningstar; B_M=21.768MHz	0ae865a8812ce93b05461ec4483ad1714e24fc9be9de1e7bb54963da43592087	**FTL.CERT**
M8P	Logical clock; BSD rank=1; CONTACT ZERO	3e5f4f044ba481fcbbb0bc731b9bbebf4adb86ec3ace716523ef4822ee64b90b	**CLOCK.CERT**
M8Q	35/35 routes GREEN; 7-ABORT matrix	81e975cf6ada9b5e9a650ecd8fcafd0b418871b2a2085ff73ac19e4aa73ceac1	**SYS.CERT**

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Full pipeline: 23-module causal DAG | All SHAs computed in environment, none fabricated
Source: github / replit.dev Opera Numerorum | Stack: Python 3.12, mpmath 1.3.0, C gcc 80-bit, reportlab 4.5.1